Solving for Regularization and Cross Validation

6

Post

CS 3244
Machine Learning

NUS National University of Singapore | Computing

Recap and Forecast

Bias: Cost due to using $\mathcal{H}$ to approximate f over any dataset.

$$\mathbb{E}_x\left(\bar{h}(x) - f(x)\right)^2$$

Variance: Cost due to the specific dataset $\mathcal{D}$.

$$\mathbb{E}_{\mathcal{D},x}\left[\left(h_{\mathcal{D}}(x) - \bar{h}(x)\right)^2\right]$$

Stochastic Noise: Uncontrollable noise.

$$\mathbb{E}_{\epsilon,x}\left(\epsilon(x)\right)^2$$

Match the 'model complexity' to the data resources, not to the target complexity.

(Soft) SVM: Optimal linear classifier determined by support vectors (with tolerance for errors)

Learning Outcomes for this week

- Understand **Regularization** as a means of restraining the model.
- Choose appropriate doses of regularization for a model.
- Understand and execute **Validation**, as a reality check by peeking (at the bottom line).
- Understand the different forms extending validation to encompass additional estimation.
- Understand how validation and regularization complement each other and their roles in affecting learning.

Solving for Regularization

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Recall: Soft Order Regularization

$$\mathcal{H}_{10} = \{h(x) = \theta_0 + \theta_1 x + \cdots + \theta_{10} x^{10}\}$$

$$\mathcal{H}_C = \{h(x) = \theta_0 + \theta_1 x + \cdots + \theta_{10} x^{10}\}$$

such that $\sum_{q=0}^{10} \theta_q^2 \leq C$

$$\mathcal{H}_2 = \{h(x) = \theta_0 + \theta_1 x + \cdots + \theta_{10} x^{10}\}$$

such that $\theta_3 = \theta_4 = \cdots = \theta_{10} = 0$

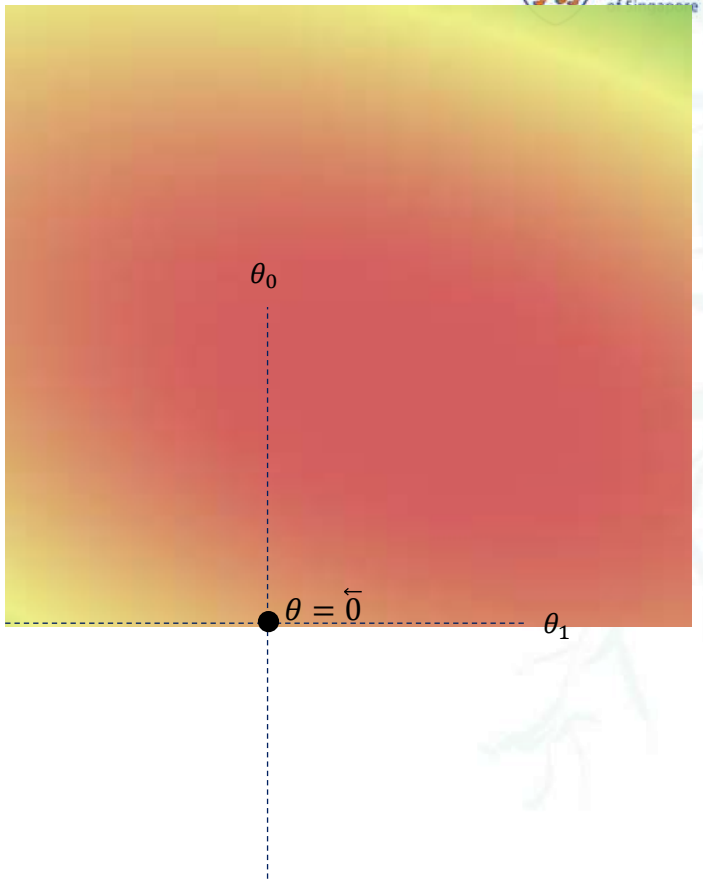
$\mathcal{H}$ is larger

$\mathcal{H}$ is smaller

Solving for θ_{reg}

Minimize $J_{train} = \frac{1}{m}(X\theta - y)^\top(X\theta - y)$

Subject to $\theta^\top\theta \leq C$



Solving for θ_{reg}

Minimize $J_{train} = \frac{1}{m}(X\theta - y)^\top(X\theta - y)$

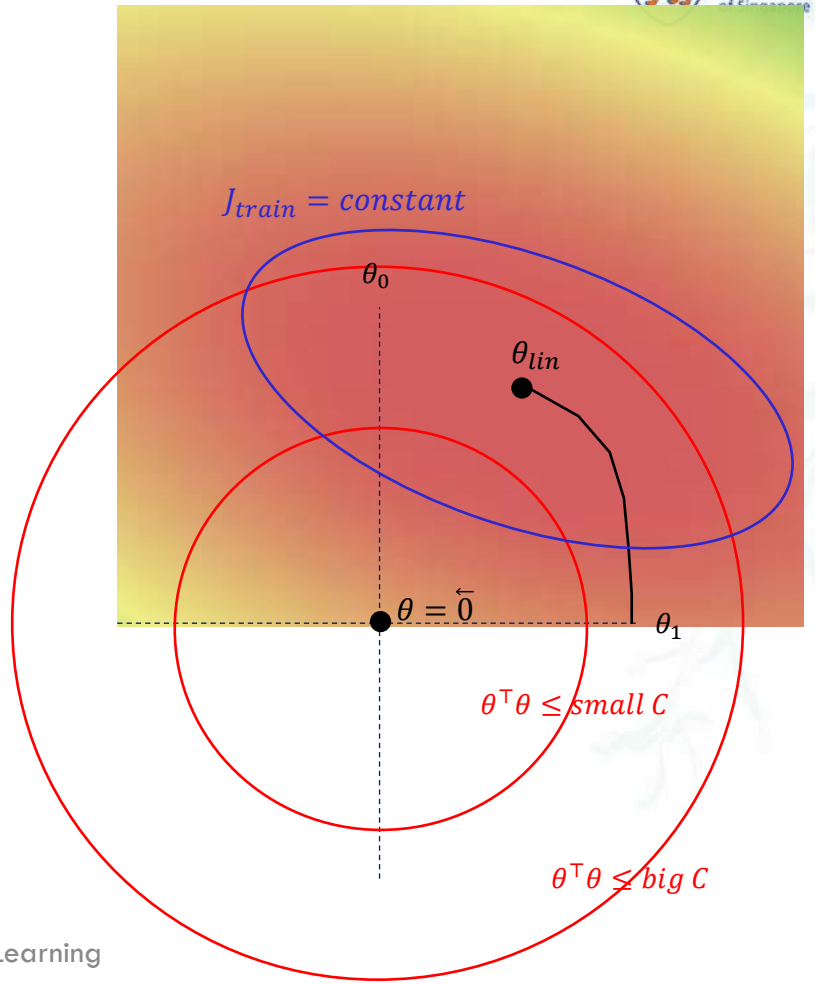
Subject to $\theta^\top\theta \leq C$

<u>Pictorially with 2 weights :</u>

J_{train} gradient as heatmap.

Blue oval is a contour where J_{train} is a constant value (same color)

Red disc defines uniform weight decay region where $\theta^\top\theta \leq C$.



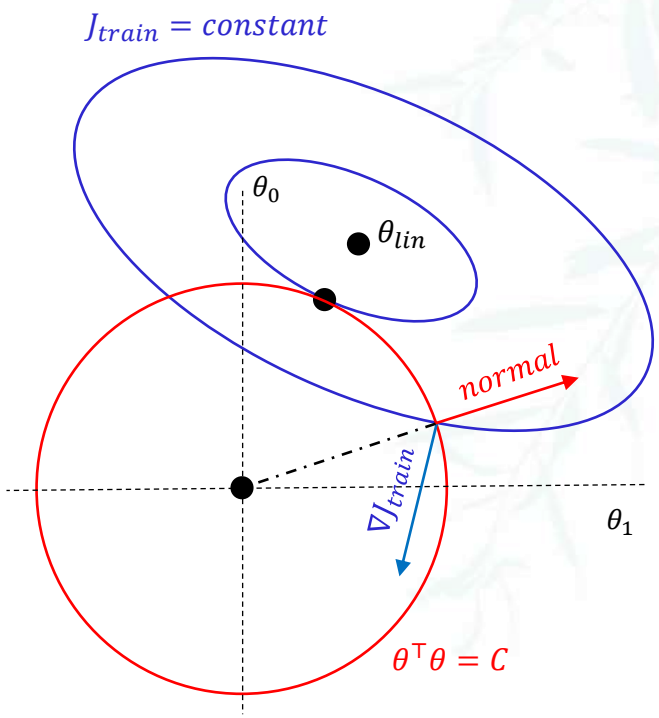
Solving for θ_{reg}

Minimize $J_{train} = \frac{1}{m}(X\theta - y)^\top (X\theta - y)$

Subject to $\theta^\top \theta \leq C$

Observations:

1. Optimal θ tries to get as 'close' to θ_{lin} as possible
   Optimal θ will use full budget and be on the surface $\theta^\top \theta = C$.
2. Surface $\theta^\top \theta = C$, at optimal θ, should be perpendicular to ∇J_{train}.
   Otherwise can move along the surface and decrease J_{train}.
3. Normal to surface $\theta^\top \theta = C$ is the vector θ.
4. Surface is $\perp$ ∇J_{train}; surface is $\perp$ normal.
   ∇J_{train} is parallel to normal (but in the opposite direction).



Solving for θ_{reg}

Minimize $J_{train} = \frac{1}{m}(X\theta - y)^\top (X\theta - y)$

Subject to $\theta^\top \theta \leq C$

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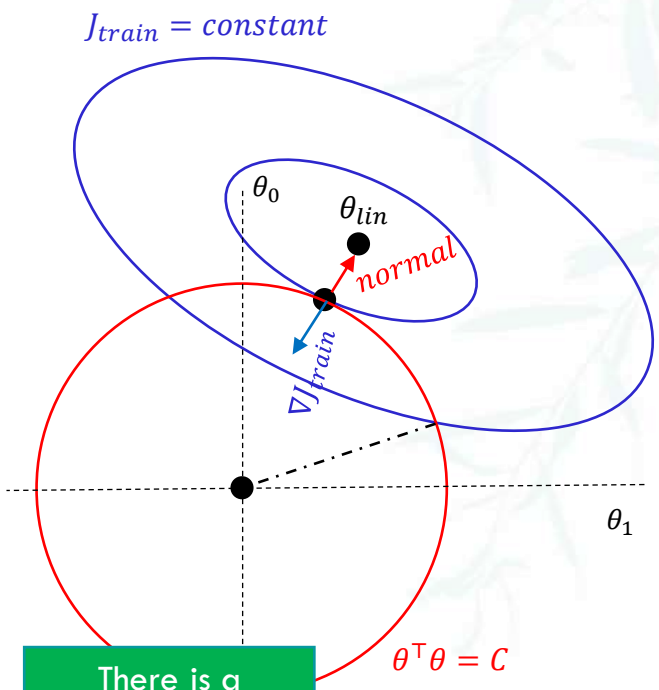
4. Surface is $\perp$ ∇J_{train}; surface is $\perp$ normal.
   ∇J_{train} is parallel to normal (but in the opposite direction).

$$\nabla J_{train}(\theta_{reg}) \propto -\theta_{reg}$$

Constant of proportionality. Chosen for convenience

$$= -2\frac{\lambda}{m}\theta_{reg}$$

There is a correspondence:
$C \uparrow \quad \lambda \downarrow$



Augmented Error

<u>Unconstrained</u>

min $J_{train} = \frac{1}{m}\sum_{j=1}^{m}\left(\theta^\top x^{(j)} - y^{(j)}\right)^2$

min $J_{train} = \frac{1}{m}(X\theta - y)(X\theta - y)$

Recall: $\nabla J_{train}(\theta_{reg}) \propto -\theta_{reg}$

$= -2\frac{\lambda}{m}\theta_{reg}$

$\theta_{lin} = (X^\top X)^{-1}Xy$

<u>Constrained:</u>

min $J_{train} = \frac{1}{m}(X\theta - y)(X\theta - y)$, subject to: $\theta^\top\theta \leq C$

Augmented Error

Unconstrained

min $J_{train} = \frac{1}{m}\sum_{j=1}^{m}\left(\theta^\top x^{(j)} - y^{(j)}\right)^2$

min $J_{train} = \frac{1}{m}(X\theta - y)(X\theta - y)$

Recall: $\nabla J_{train}(\theta_{reg}) \propto -\theta_{reg}$

$= -2\frac{\lambda}{m}\theta_{reg}$

$\theta_{lin} = (X^\top X)^{-1}Xy$

Constrained:

min $J_{train} = \frac{1}{m}(X\theta - y)(X\theta - y)$, subject to: $\theta^\top\theta \leq C$

$\equiv$

min $J_{aug}(\theta) = J_{train}(\theta) + \frac{\lambda}{m}\theta^\top\theta$

$= \frac{1}{m}\left((X\theta - y)^\top(X\theta - y) + \lambda\theta^\top\theta\right)$

Compare versus implicit constraint

Take derivatives:

Set $\nabla J_{aug}(\theta) = \vec{0} \Longrightarrow$

$= X^\top(X\theta - y) + \lambda\theta$

$= (X^\top X + \lambda I)\theta - X^\top y$

$\theta_{reg} = (X^\top X + \lambda I)^{-1}X^\top y$

Still 1 step learning!
λ determines the amount of regularization.

Regularization Variants

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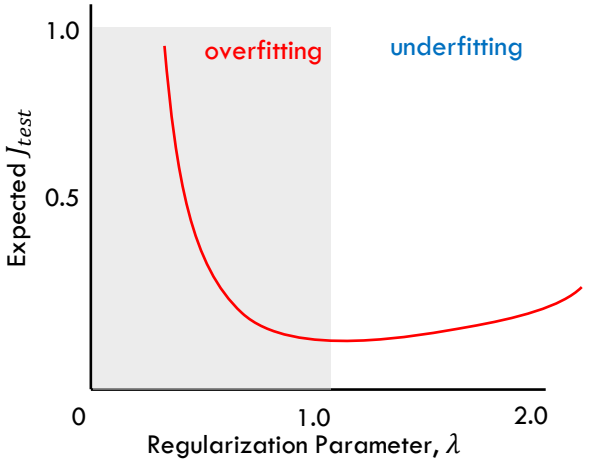
Regularizer variations $\Omega(h)$

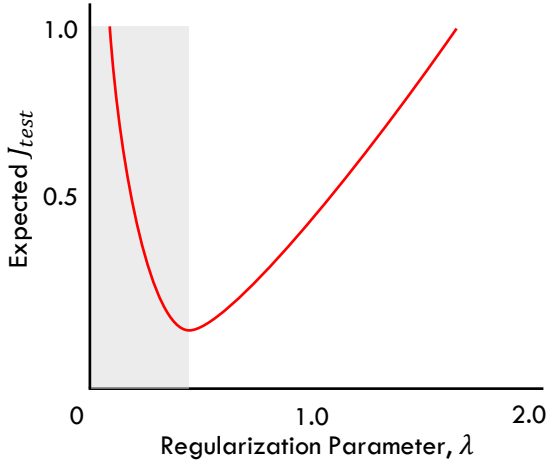
Why is it called "weight decay"? We'll see in neural networks.

Uniform weight decay



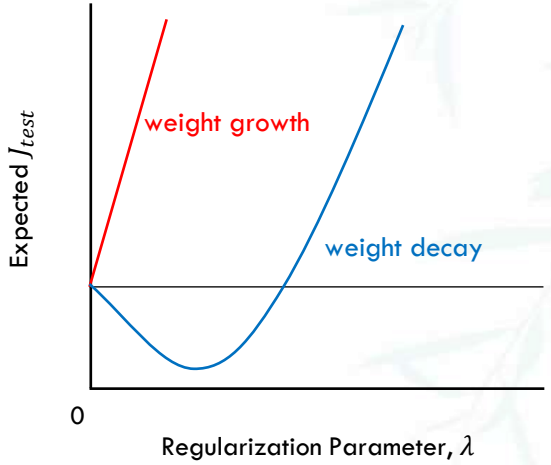
$$\sum_{q=0}^{Q} \theta_q^2$$

Low Order Fit



$$\sum_{q=0}^{Q} q\theta_q^2$$

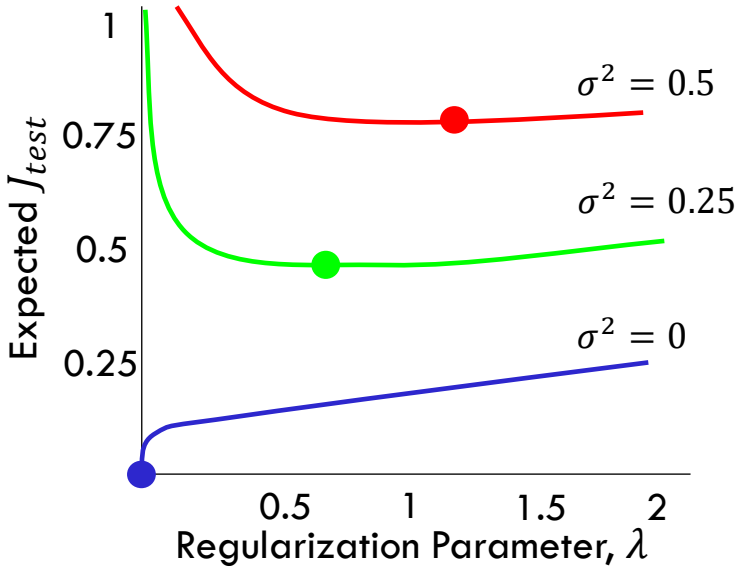
Weight growth

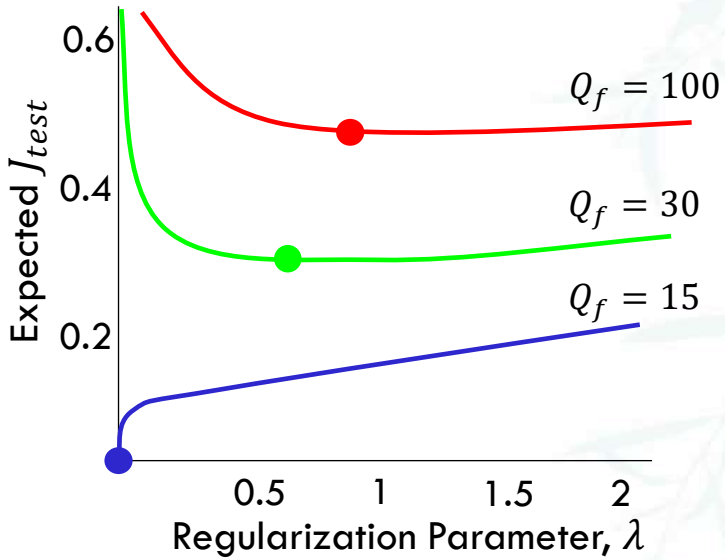


$$\sum_{q=0}^{Q} \frac{1}{\theta_q^2}$$

Getting the right dose

Recall the bias-variance experiment varying σ^2 (noise) and Q_f (target complexity) :





Stochastic Noise

(high frequency)

Deterministic Noise (Bias)

(also non-smooth w.r.t. to $\mathcal{H}$)

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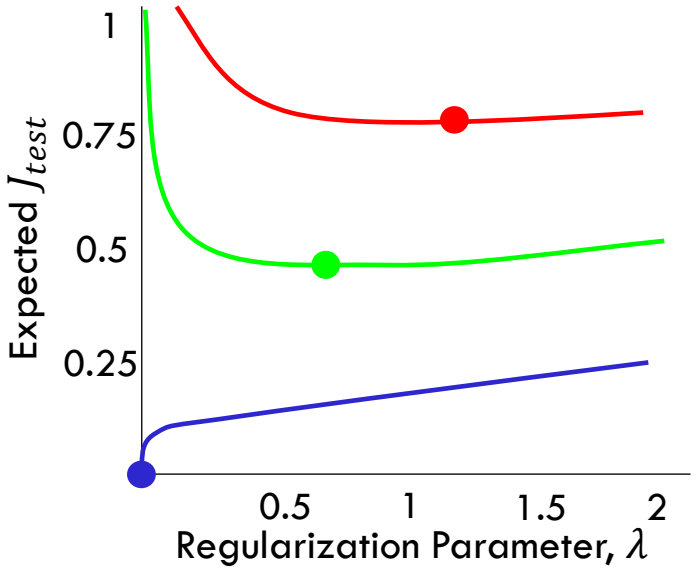
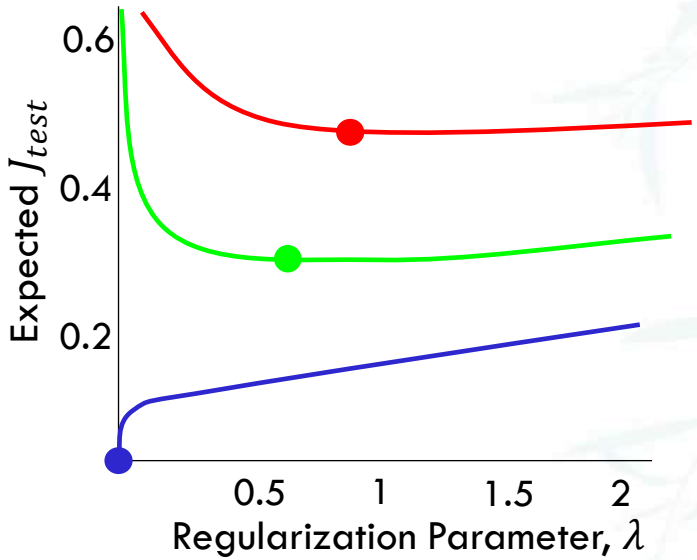
Recall the bias-variance experiment varying σ^2 (noise) and Q_f (target complexity) :



Stochastic Noise

(high frequency)



Deterministic Noise (Bias)

(also non-smooth w.r.t. to $\mathcal{H}$)

The perfect regularizer Ω

Constraint in the 'direction' of the target function

- But we don't know f: circular argument!

Guiding principle:

- Direction of a smoother or "simpler" hypothesis
- Smoother = impairs our ability to fit (high-frequency) noise
- Sacrifice a little bias for large improvement on variance

Chose a bad Ω?

- We still have λ! $\Longrightarrow$ validation, up next

Regularization – Summary

Give up modeling a subset of of $\mathcal{H}$ to lower variance error.

$$J_{aug}(\theta) = J_{train}(\theta) + \frac{\lambda}{m}\Omega(h)$$

λ is the dose of regularization:

the higher the dose, the smaller the budget C.

Choosing a regularizer $\Omega(h)$:

Weight decay is common $\sum_{q=0}^{Q} \theta_q^2$ (L2 regularization)

Choose λ by validation …

Cross Validation

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The dilemma about K

Validation relies on the following chain of reasoning:

True when K is big

$$J_{test}(h) \approx J_{test}(h^{-}) \approx J_{val}(h^{-})$$

True when K is small

Can we have both K being both big and small?

Yes, we can!

Leave out out cross validation

$m - 1$ points for training and 1 point for validation
(Sounds familiar? It was at the beginning of the pre validation lecture)

$\mathcal{D}_{cv} = \left(x^{(1)}, y^{(1)}\right), \ldots, \cancel{\left(x^{(cv)}, y^{(cv)}\right)}, \ldots, \left(x^{(m)}, y^{(m)}\right)$ validation

Final hypothesis learned from $\mathcal{D}_{cv}$ is h_{cv}^{-}.

$$j_{cv} = j_{val}(h_{cv}^{-}) = j\left(h_{cv}^{-}\left(x^{(cv)}\right), y^{(cv)}\right)$$

Caveat: Hypothesis learned will be highly correlated.
The first hypothesis will use points 2, 3, …, m and the second will use 1, 3, … , m, as most points are the same.

Leave out out cross validation

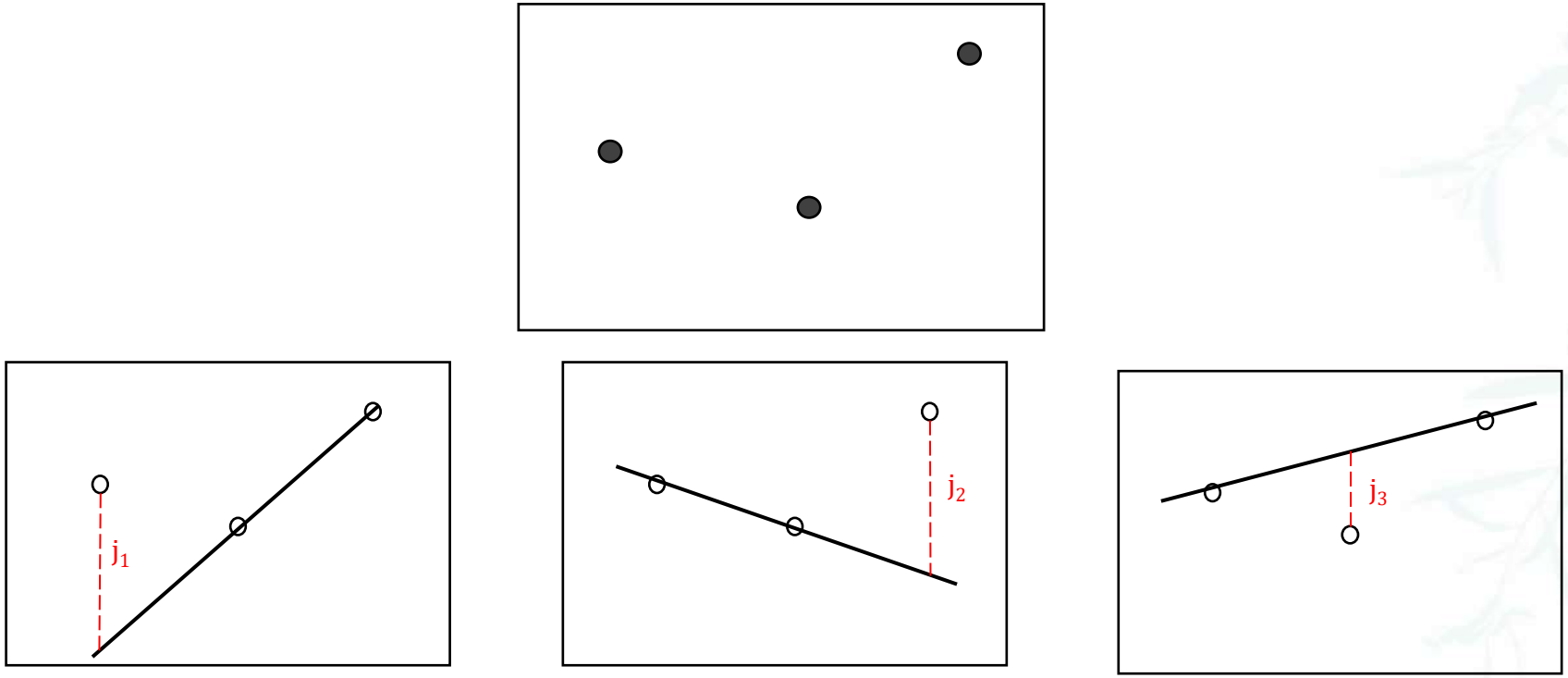
Final hypothesis learned from $\mathcal{D}_{cv}$ is h_{cv}^{-}.

$$j_{cv} = j_{val}(h_{cv}^{-}) = j\left(h_{cv}^{-}\left(x^{(cv)}\right), y^{(cv)}\right)$$

Cross validation cost: $J_{loocv} = \frac{1}{m}\sum_{cv=1}^{m} j_{cv}$.

(Almost) using m examples for K.

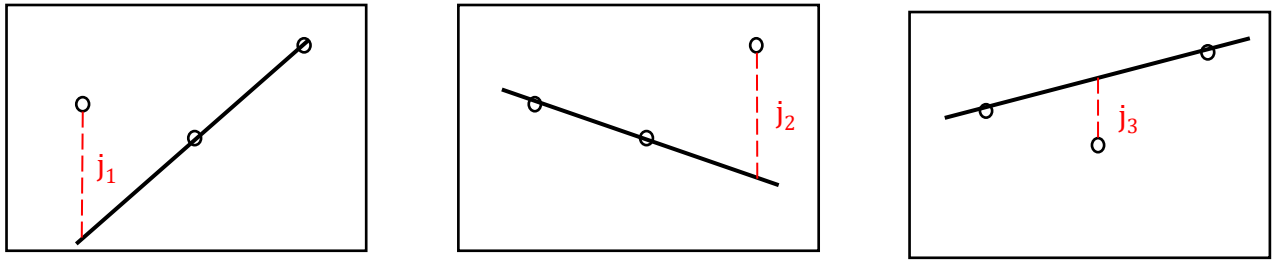
Illustration of cross validation



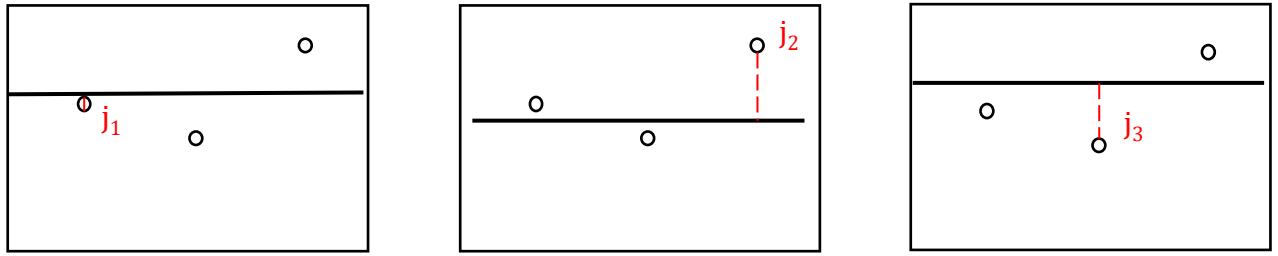
$$J_{loocv} = \frac{1}{3}(j_1 + j_2 + j_3)$$

Model selection using CV

Linear

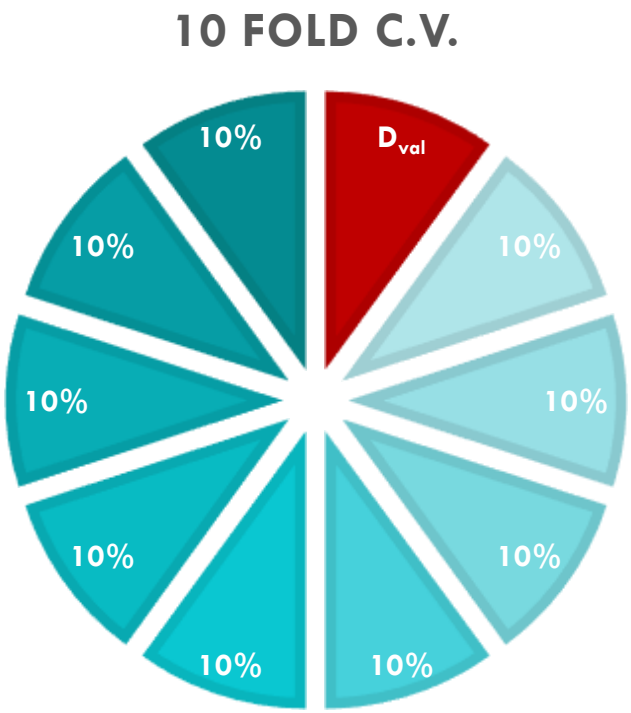


Constant



J_{cv} empirically shows that the constant model is a better fit for this dataset

Have your cake and eat it too: K fold cross validation



LOOCV can be very expensive for large datasets. Why?

Instead, use K fold cross validation: i.e., K training sessions on $\frac{m}{K}$ points each.

Recommend: 10-fold CV

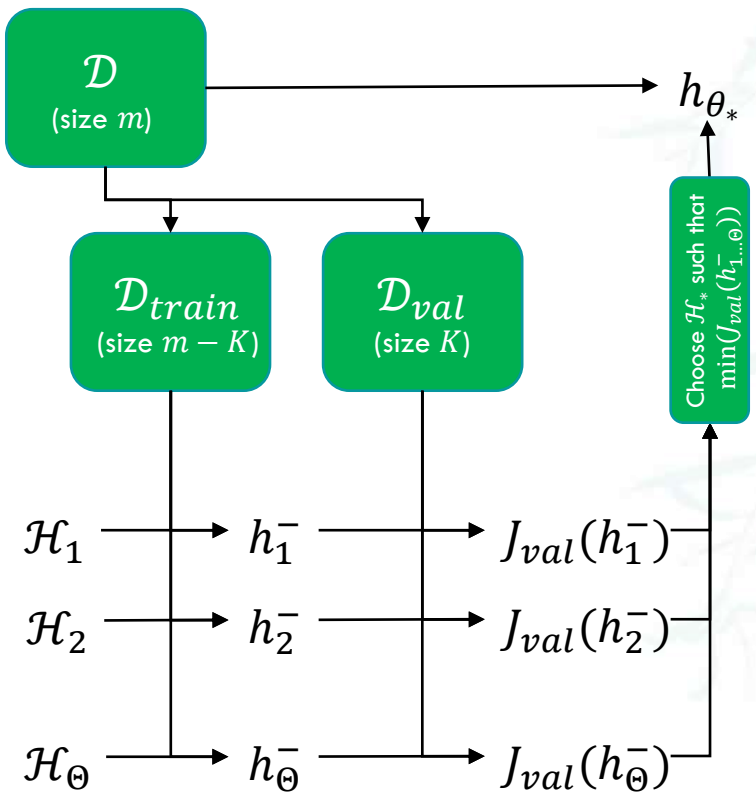
Cross Validation – Summary

Estimate J_{val} multiple times

Breaks independence assumption: performance is correlated.

Introduces the factor of efficiency as a tradeoff.

To think about: How else can we produce estimates of J_{test}?



Validation Case Study

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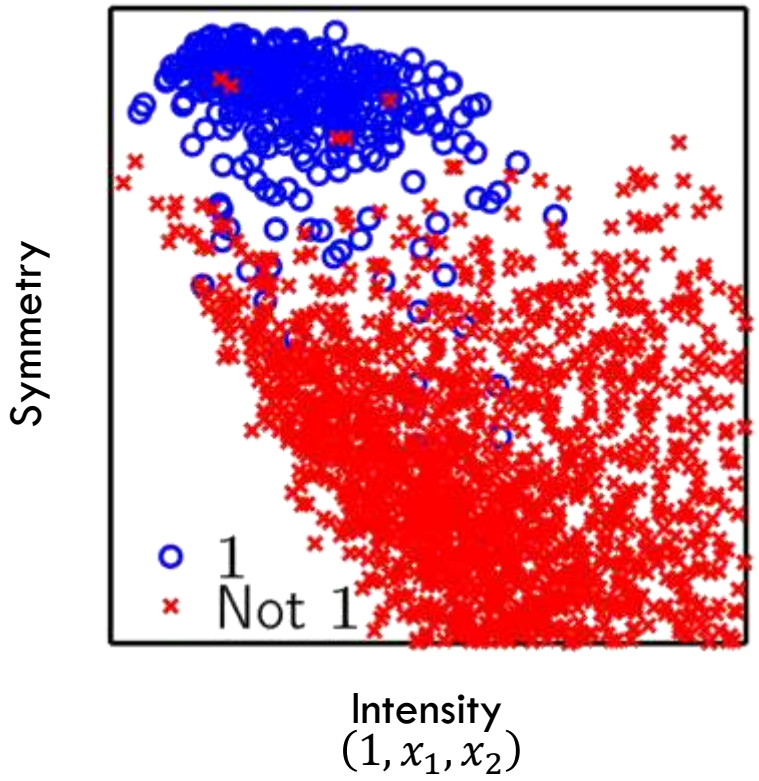
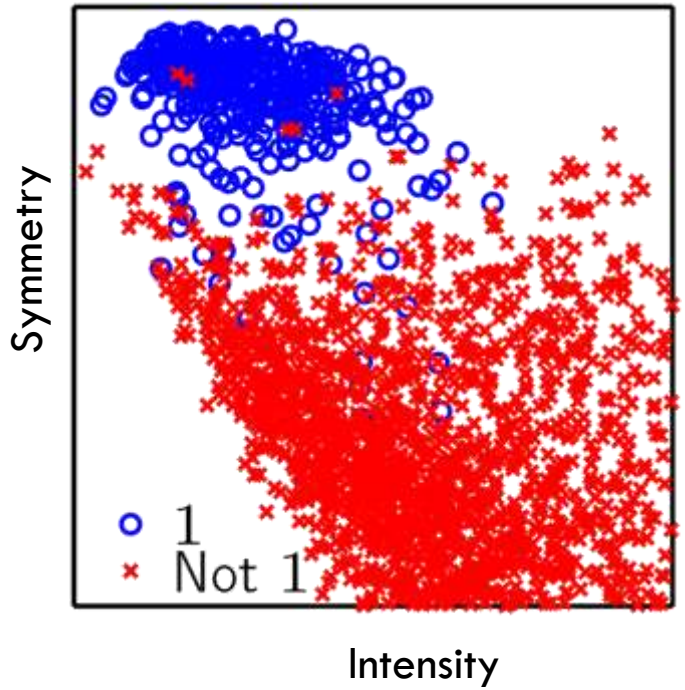
CV on MNIST

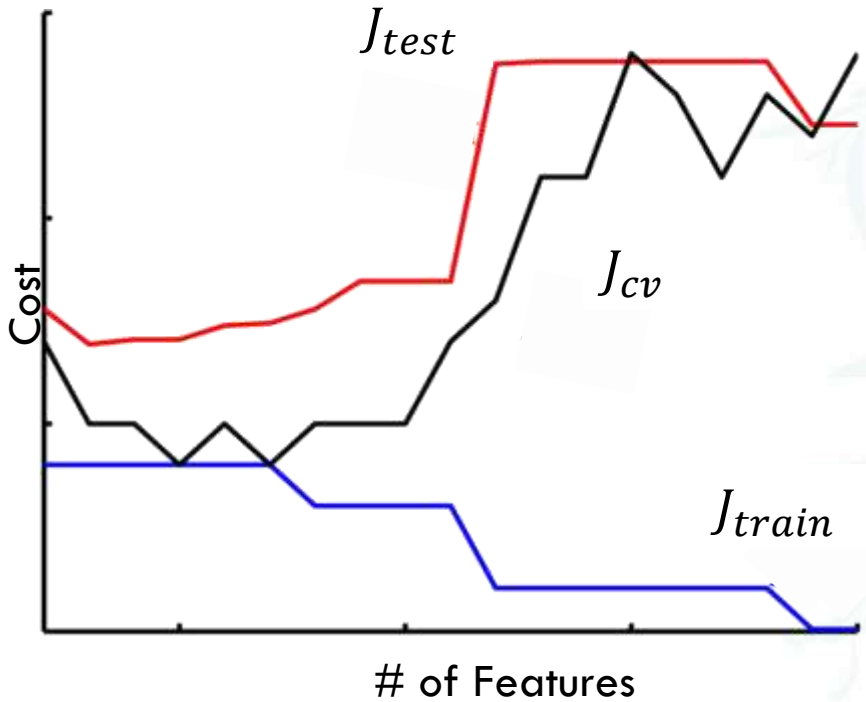


Initially we have only
x_1 : symmetry and
x_2 : intensity

Limited to a 1D hyperplane

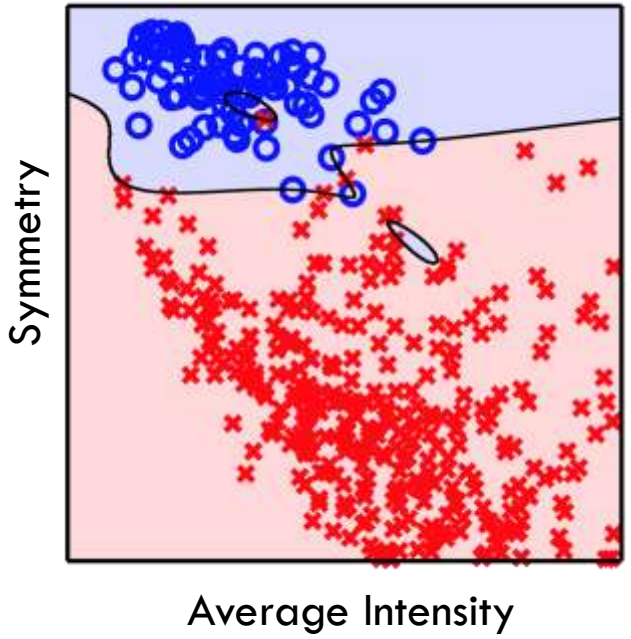
CV on MNIST





$$(1, x_1, x_2) \rightarrow \left(1, x_1, x_2, x_1^2, x_1x_2, x_2^2, x_1^3, x_1^2x_2, x_1x_2^2, \ldots, x_1x_2^4, x_2^5\right)$$

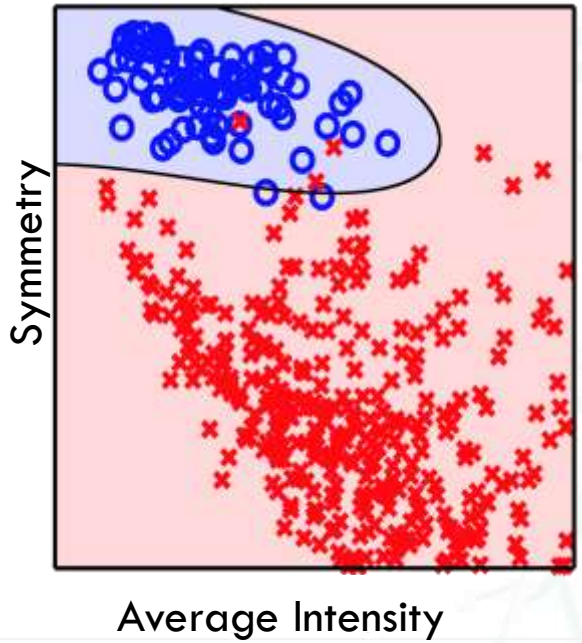
Validation Wins in the Real World



No validation (20 features)

$J_{train} = 0\%$

$J_{test} = 2.5\%$



Leave one out cross validation (6 features)

$J_{train} = 0.8\%$

$J_{test} = 1.5\%$

Huge improvement of 40% error reduction

Regularizing with Lp norms

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Geometry of ℓ^p norms

$$||x||_p \equiv \left(\sum_{i=1}^{n} |x_i|^p\right)^{\frac{1}{p}}$$

$\ell 1$ = Manhattan (Taxicab) distance

$\sum_{i=1}^{n} |x_i|$

$\ell 2$ = Euclidean distance (weight decay)

$\sqrt[2]{\sum_{i=1}^{n} |x_i|^2}$

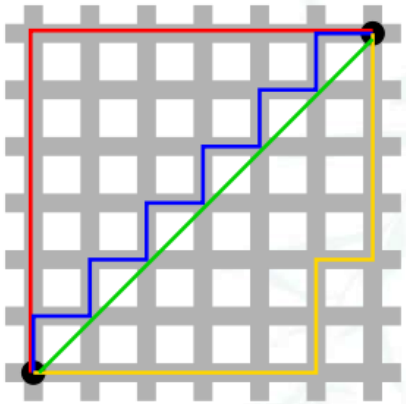


Diagram Credit: Psychonaut via Wikipedia, public domain

ℓ^p ball visualized

As the value of p decreases, the size of the corresponding space also decreases.

In 3 equally weighted dimensions (e.g., x_1, x_2, x_3):

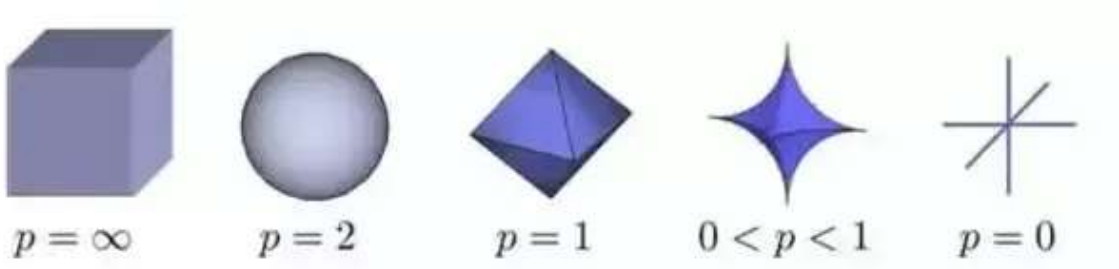
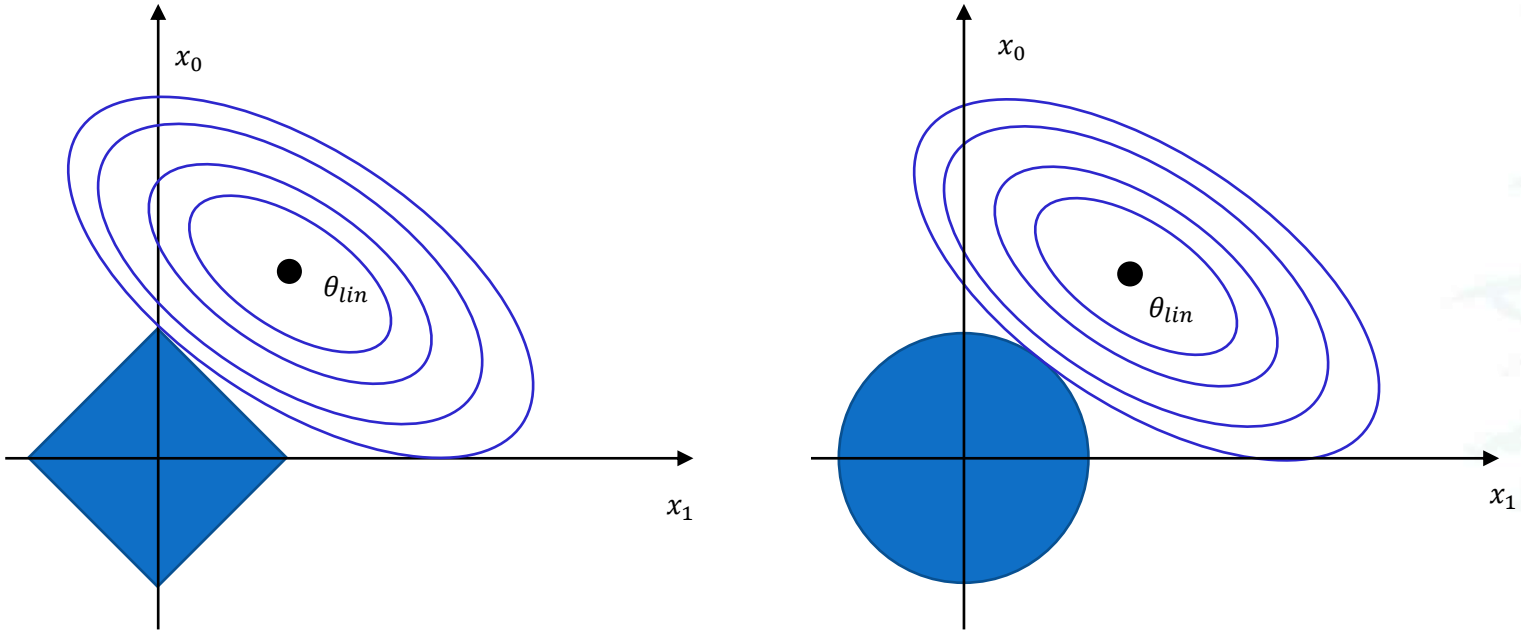


Diagram Credit: Manjari Narayan via Quora.

Properties from geometry

- L1 is said to encourage sparse solutions; feature selection
- L2 can be used for homogenous data